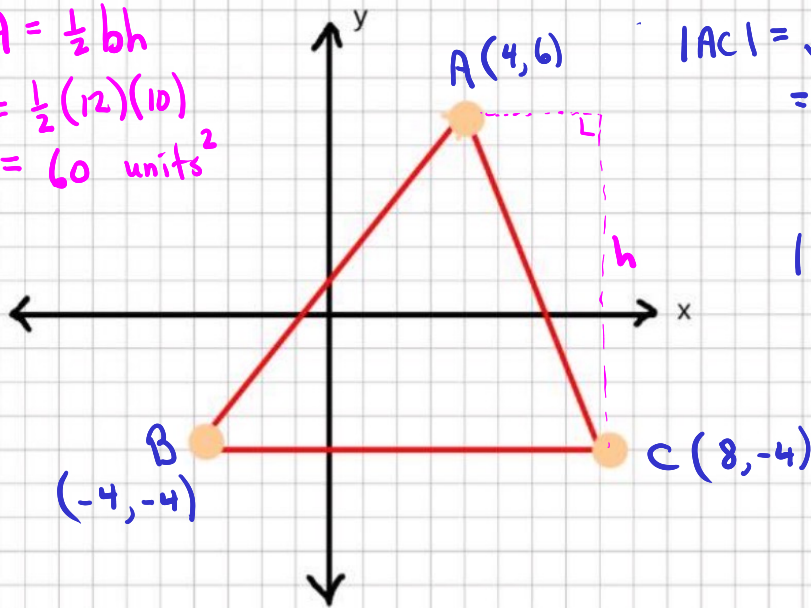


3. Determine the distance between the points (2, 7) and (3, 9).

4. Determine the perimeter and the area of the following triangle.

$$\begin{aligned} A &= \frac{1}{2}bh \\ &= \frac{1}{2}(12)(10) \\ &= 60 \text{ units}^2 \end{aligned}$$



$$\begin{aligned} |AC| &= \sqrt{(8-4)^2 + (-4-6)^2} \\ &= \sqrt{16 + 100} \\ &= 10.8 \end{aligned}$$

$$\begin{aligned} |AB| &= \sqrt{(-4-4)^2 + (-4-6)^2} \\ &= \sqrt{64 + 100} \\ &= 12.8 \end{aligned}$$

$$\begin{aligned} |BC| &= \sqrt{(8-(-4))^2 + (-4-(-4))^2} \\ &= \sqrt{144 + 0} \\ &= 12 \end{aligned}$$

$$\begin{aligned} \therefore P &= 10.8 + 12.8 + 12 \\ &= 35.6 \text{ units} \end{aligned}$$

5. Solve the following system of equations using a graph:

$$\begin{aligned} y &= -2x + 3 \\ y &= 3x - 5 \end{aligned}$$